

$$h(x) = h(x_0) + \frac{h'(x_0)}{1!}(x-x_0) + \frac{h''(x_0)}{2!}(x-x_0)^2 + \dots + \frac{h^{(n)}(x_0)}{n!}(x-x_0)^n$$

a) Compute all Taylor polynomials of $f(x) = x^4 + 2x^3 + x^2 + 5$ around $x_0 = 1$.

$$f(x) = x^4 + 2x^3 + x^2 + 5 \Rightarrow f(1) = 9$$

$$f'(x) = 4x^3 + 6x^2 + 2x \Rightarrow f'(1) = 12$$

$$f''(x) = 12x^2 + 12x + 2 \Rightarrow f''(1) = 36$$

$$f^{(3)}(x) = 24x + 12 \Rightarrow f^{(3)}(1) = 36$$

$$f^{(4)}(x) = 24 \Rightarrow f^{(4)}(1) = 24$$

$$f^{(5)}(x) = 0 \Rightarrow f^{(5)}(1) = 0$$

$$T_1(x) = 9$$

$$T_2(x) = 9 + 12(x-1)$$

$$T_3(x) = 9 + 12(x-1) + 18(x-1)^2$$

$$T_4(x) = 9 + 12(x-1) + 18(x-1)^2 + 6(x-1)^3$$

$$T_5(x) = 9 + 12(x-1) + 18(x-1)^2 + 6(x-1)^3 + 2(x-1)^4$$

b) Compute the Taylor series of $g(x) = \ln(1+x)$ around $x_0 = 0$.

$$g(x) = \ln(1+x) \Rightarrow g(0) = 0$$

$$g'(x) = \frac{1}{1+x} \Rightarrow g'(0) = 1$$

$$g''(x) = -\frac{1}{(1+x)^2} \Rightarrow g''(0) = -1$$

$$g^{(3)}(x) = \frac{2}{(1+x)^3} \Rightarrow g^{(3)}(0) = 2$$

$$g^{(4)}(x) = -\frac{6}{(1+x)^4} \Rightarrow g^{(4)}(0) = -6$$

$$g^{(5)}(x) = \frac{24}{(1+x)^5} \Rightarrow g^{(5)}(0) = 24$$

$$\vdots$$

$$g(x) = 0 + x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 + \dots \Rightarrow \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$$

Show that for a function $f \in C^L([0,1];\mathbb{R})$, we have the following expression

for the remainder of the Taylor expansion (about 0, evaluated at 1),

$$f(1) - \sum_{k=0}^{m-1} \frac{f^{(k)}(0)}{k!} = \int_0^1 \frac{1}{m!} s^{m-1} f^{(m)}(s-1) ds$$

for every $m \leq L$.

Idk?

We want to find a difference formula of the form

$$u''(x) \approx \frac{a_1 u(x) + a_2 u(x-h) + a_3 u(x-\frac{1}{2}h)}{h^2}$$

where a_1 , a_2 and a_3 are constants to be determined.

Find the coefficients a_1 , a_2 and a_3 which makes this scheme convergent.

Find an expression for the error term.

We know that the sum of $u(x)$, the sum of $u'(x)$ needs to be zero and sum of $u''(x)$ needs to be one

Then we have three equations we can use to calculate a_i .

$$u''(x) = \frac{a_1 u(x) + a_2 u(x-h) + a_3 u(x-\frac{1}{2}h)}{h^2}$$

First, we expand $u(x-h)$ and $u(x-\frac{1}{2}h)$ to the power of three

$$u''(x) = \frac{a_1 u(x) + a_2 u(x-h) + a_3 u(x-\frac{1}{2}h)}{h^2}$$

$$u(x-h) = u(x) - u'(x)h + \frac{u''(x)h^2}{2} - \frac{u^{(3)}(x)h^3}{6} + O(h^4)$$

$$u(x-\frac{1}{2}h) = u(x) - \frac{u'(x)h}{2} + \frac{u''(x)h^2}{8} - \frac{u^{(3)}(x)h^3}{48} + O(h^4)$$

After we substitute and calculate, we're left with the expression for $u''(x)$

$$u''(x) = \frac{a_1 u + a_2 u - a_2 u' h + \frac{1}{2} a_2 u'' h^2 - \frac{1}{6} a_2 u^{(3)} h^3 + a_3 u - \frac{1}{2} a_3 u' h + \frac{1}{8} a_3 u'' h^2 - \frac{1}{48} a_3 u^{(3)} h^3}{h^2}$$

$$= \frac{a_1 u + a_2 u + a_3 u - a_2 u' h + \frac{1}{2} a_3 u' h + \frac{1}{2} a_2 u'' h^2 + \frac{1}{8} a_3 u'' h^2 + O(h^3)}{h^2}$$

Now we can put our equations in a matrix and find our constants

$$\begin{bmatrix} 1 & 1 & 1 & | & 0 \\ 0 & -1 & -\frac{1}{2} & | & 0 \\ 0 & \frac{1}{2} & \frac{1}{8} & | & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & | & 4 \\ 0 & 0 & 1 & | & -8 \\ 0 & 1 & 0 & | & 4 \end{bmatrix} \quad \begin{array}{l} a_1 = 4 \\ a_2 = 4 \\ a_3 = -8 \end{array}$$

After putting the constants into the eq., we find an expression for the error.

$$u''(x) = \frac{u'' h^2 - \frac{1}{2} u^{(3)} h^3}{h^2} \Rightarrow u'' = u'' - \frac{h}{2} u^{(3)} \Rightarrow e(h) = \frac{h}{2} u^{(3)}(\xi)$$

Given the two point boundary value problem

$$u_{xx} + 2u_x + \pi^2 u = \cos(\pi x) - \pi(x+1)\sin(\pi x), \quad 0 \leq x \leq 2$$

$$u(0) = 0, \quad u(2) = 1$$

a) Verify that the exact solution is

$$u(x) = \frac{x}{2} \cos(\pi x).$$

$$u'' + p(x)u' + q(x)u = r(x)$$

$$p(x) = 2, \quad q(x) = \pi^2, \quad r(x) = \cos(\pi x) - \pi(x+1)\sin(\pi x)$$

First we calculate the first and second derivative of the given solution;

$$u'(x) = \frac{1}{2} \cos(\pi x) - \frac{\pi x}{2} \sin(\pi x)$$

$$u''(x) = -\frac{\pi}{2} \sin(\pi x) - \frac{\pi}{2} \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x) - \pi \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x)$$

Then we use our derivatives to check our original equation:

$$\begin{aligned} & \left(-\pi \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x) \right) + 2 \left(\frac{1}{2} \cos(\pi x) - \frac{\pi x}{2} \sin(\pi x) \right) + \pi^2 \left(\frac{x}{2} \cos(\pi x) \right) = r(x) \\ & = -\pi \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x) + \cos(\pi x) - \pi x \sin(\pi x) + \frac{\pi^2 x}{2} \cos(\pi x) \\ & = \cos(\pi x) - \pi(1+x)\sin(\pi x) = r(x) \end{aligned}$$

$$\begin{aligned} \text{b)} \quad & \frac{u(x_i+h) - 2u(x_i) + u(x_i-h))}{h^2} + p(x_i) \frac{u(x_i+h) - u(x_i-h))}{2h} + q(x_i) u(x_i) + O(h^2) = r(x) \\ & \frac{u(x_i+h) - 2u(x_i) + u(x_i-h))}{h^2} + 2 \frac{u(x_i+h) - u(x_i-h))}{2h} + \pi^2 u(x_i) = \cos(\pi x) - \pi(x+1)\sin(\pi x) \end{aligned}$$

$$U_i \approx u(x_i), \quad u(x_i \pm h) \approx U_{i \pm 1}$$

$$\begin{aligned} & \frac{U_{i+1}}{h^2} - \frac{2U_i}{h^2} + \frac{U_{i-1}}{h^2} + \frac{U_{i+1}}{h} - \frac{U_{i-1}}{h} + \pi^2 U_i = r(x) \\ & \underbrace{\frac{U_{i+1}}{h^2} - \frac{2U_i}{h^2} + \frac{U_{i-1}}{h^2}}_{a_1} + \underbrace{\frac{U_{i+1}}{h} - \frac{U_{i-1}}{h}}_{a_2} + \underbrace{\pi^2 U_i}_b = \underbrace{r(x)}_c \end{aligned}$$

$$U_{i-1}(1-h) + U_i(\pi^2 - 2) + U_{i+1}(1+h) = h^2 r(x)$$

$$c) \quad N=4 \Rightarrow h = \frac{2-0}{4} = 0.5 \Rightarrow i = 0, 0.5, 1, 1.5, 2$$

$$\frac{u(x_i+h) - 2u(x_i) + u(x_i-h))}{h^2} + p(x_i) \frac{u(x_i+h) - u(x_i-h))}{2h} + q(x_i)u(x_i) + O(h^2) = r(x)$$

$$\frac{u(x_i+h) - 2u(x_i) + u(x_i-h))}{h^2} + 2 \frac{u(x_i+h) - u(x_i-h))}{2h} + \alpha^2 u(x_i) = \cos(\alpha x) - \alpha(x+1)\sin(\alpha x)$$

$$U_i \approx u(x_i), \quad u(x_i \pm h) \approx U_{i \pm 1}$$

$$\frac{U_{i+1}}{h^2} - \frac{2U_i}{h^2} + \frac{U_{i-1}}{h^2} + \frac{U_{i+1}}{h} - \frac{U_{i-1}}{h} + \alpha^2 U_i = r(x)$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ a_{1,1} & a_{2,1} & a_{3,1} & 0 & 0 \\ 0 & a_{1,2} & a_{2,2} & a_{3,2} & 0 \\ 0 & 0 & a_{1,3} & a_{2,3} & a_{3,3} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} U_0 \\ U_1 \\ U_2 \\ U_3 \\ U_4 \end{bmatrix}$$

$$b = \begin{bmatrix} h^2 r(x_0) \\ h^2 r(x_1) \\ h^2 r(x_2) \\ h^2 r(x_3) \\ h^2 r(x_4) \end{bmatrix}$$

After a lot of calculations, we get:

$$AU = \vec{b} \Rightarrow U_0 = 0 \quad U_1 = -3,0791 \quad U_2 = 0,1740 \quad U_3 = 0,8054 \quad U_4 = 1$$